

DATA QUALITY ASSESSMENT

Data Quality Assessment has been performed on data provided by Food-Junction chain of restaurants. SQL tools have been used to perform data profiling which revealed that data is well organized and clean. There are no missing fields or null values.

Exploratory Data Analysis

- Dataset contains 20,000 orders across 8 branches.
- Order activity spans January 2024–December 2025.
- Pakistani cuisine accounts for the largest menu category.
- Customer base consists of 5,000 unique customers.
- Data distribution appears suitable for further analysis.

The size of data provided over two years of time can be used to explore meaningful patterns. Data is also consistent and complete. The formats of each data type are also consistently correct. The errors or missing data found so far is negligible, therefore, there are no calculations to predict the incorrect impacts on the assessment.

Query Results Summarized in Tables:

1. Total Rows per Table		2. Duplicate Counts		3. Primary Key(PK) Uniqueness		
Table	Total Records	Table	Duplicate Records	Table	PK	Result
branches	8	branches	0	branches	Branch_ID	Pass
employees	112	employees	0	employees	Employee_ID	Pass
customers	5,000	customers	0	customers	Customer_ID	Pass
menu_items	80	orders	0	orders	Order_ID	Pass
orders	20,000					
order_details	N/A					
reviews	N/A					

4. Foreign Key (FK) Integrity			5. Date Range			6. Numeric Validation		
Relationship		Result	Column	Min	Max	Column	Check	Result
Employee	→ Branch	Pass				Quantity	1–3	Pass
Orders	→ Customer	Pass				Price	>0	Pass
Orders	→ Branch	Pass	Order_Date	2024-01-01	2025-12-31	Cost	>0	Pass
Order Details	→ Orders	Pass	Hire_Date	2024-01-02	2025-11-15	Rating	1–5	Pass
Order Details	→ Menu Items	Pass						
Reviews	→ Orders	Pass						

Table Notes:

1. Date size including number of tables and their rows.
2. No duplicate records were found.
3. All Primary keys have unique entries in each of their rows.
4. Foreign keys are connected in relation to their parent table. No orphan records were found.
5. Order and hire dates are well within the dates of date and time range of data.
6. Numbers are also in right range and pattern based on the value they are representing.

Appendices

SQL commands and their outcomes are attached.

PK uniqueness query (example duplicates):

```
SELECT
    Customer_ID,
    COUNT(*) AS Duplicate_Count
FROM customers
GROUP BY Customer_ID
HAVING COUNT(*) > 1;
```

FK integrity query:

orders -- duplicates

```
SELECT o.*
FROM orders o
LEFT JOIN customers c
ON o.Customer_ID = c.Customer_ID
WHERE c.Customer_ID IS NULL;
```

Orders -- branches

```
SELECT o.*
FROM orders o
LEFT JOIN branches b
ON o.Branch_ID = b.Branch_ID
WHERE b.Branch_ID IS NULL;
```

Employees – branches

```
SELECT e.*
FROM employees e
LEFT JOIN branches b
ON e.Branch_ID = b.Branch_ID
```

WHERE b.Branch_ID IS NULL;

Order Details – Orders

```
SELECT od.*  
FROM order_details od  
LEFT JOIN orders o  
ON od.Order_ID = o.Order_ID  
WHERE o.Order_ID IS NULL;
```

Order Details – Menu items

```
SELECT od.*  
FROM order_details od  
LEFT JOIN menu_items m  
ON od.Menu_Item_ID = m.Menu_Item_ID  
WHERE m.Menu_Item_ID IS NULL;
```

Reviews – Orders

```
SELECT r.*  
FROM reviews r  
LEFT JOIN orders o  
ON r.Order_ID = o.Order_ID  
WHERE o.Order_ID IS NULL;
```

SCHEMAS

Filter objects

- company_tute
 - food_junction**
 - Tables
 - branches
 - customers
 - employees
 - menu_items
 - order_details
 - orders
 - reviews
 - Views
 - Stored Procedures
 - Functions
 - parks_and_recreation
 - portfolio_sol
- Administration
- Schemas

Information:

Column:
Selling_Price_PKR

Definition:
Selling_Price_PKR int

Object Info Session

Limit to 50000 rows

```

1 • USE food_junction;
2
3 -- 0 Checking for Data Quality
4
5 • select * from customers;
6 • select * from orders;
7 • select * from employees;
8 • select * from menu_items;
9 • select * from order_details;
10 • select * from orders;

```

Result Grid

	Customer_ID	First_Name	Last_Name	Gender	City	Join_Date
1	1	Saad	Shah	M	Sialkot	2025-03-15
2	2	Maryam	Aslam	M	Faisalabad	2024-08-20
3	3	Abdullah	Chaudhry	F	Rawalpindi	2025-10-28
4	4	Ahmed	Ahmed	M	Lahore	2024-02-13
5	5	Noor	Iqbal	M	Faisalabad	2025-03-25
6	6	Abdullah	Malik	F	Faisalabad	2024-07-03
7	7	Ahsan	Khan	M	Lahore	2025-03-10
8	8	Abdullah	Qureshi	M	Lahore	2024-11-22

customers 53 x

Output

Action Output

#	Time	Action	Message
110	14:14:33	SELECT DISTINCT Selling_Price_PKR from menu_items order by Selling_Price_PKR LIMIT ...	30 row(s) returned

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SQL File 3* x data_quality queries

Limit to 50000 rows

```

7 • select * from employees;
8 • select * from menu_items;
9 • select * from order_details;
10 • select * from orders;
11
12 • select Order_id, Branch_ID , Order_DateTime from orders;
13
14 -- After having a general glance at data, data profiling is proceeded wi
15 -- missing values, nulls and duplicates
16

```

Result Grid

	Order_Detail_ID	Order_ID	Menu_Item_ID	Quantity
1	1	1	78	1
2	2	1	37	2
3	3	2	57	3
4	4	2	47	1
5	5	3	56	2
6	6	3	39	1
7	7	4	52	3
8	8	4	54	1

order_details 54 x

Output

Action Output

#	Time	Action	Message
111	14:41:22	select * from customers LIMIT 0, 50000	5000 row(s) returned

Local instance MySQL80 x

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Navigator SQL File 3* data_quality queries Limit to 50000 rows

SCHEMAS

Filter objects

- company_tute
- food_junction
 - branches
 - customers
 - employees
 - menu_items
 - order_details
 - orders
 - reviews
- views
- Stored Procedures
- Functions
- parks_and_recreation
- northfolk_end
- Administration Schemas

Column:
Selling_Price_PKR

Definition:
Selling_Price_PKR int

```

13
14 -- After having a general glance at data, data profiling is proceeded with checking out
15 -- missing values, nulls and duplicates
16
17 -- 1.a applying CASE queries
18 • SELECT * from customers;
19 • SELECT
20   SUM(CASE WHEN customers.first_name IS NULL OR customers.first_name = '' THEN 1 ELSE 0 END) AS cust_null,
21   SUM(CASE WHEN customers.Gender IS NULL OR customers.Gender = '' THEN 1 ELSE 0 END) AS gender_null,
22   SUM(CASE WHEN customers.city IS NULL OR customers.city = '' THEN 1 ELSE 0 END) AS city_null,

```

Result Grid Filter Rows: Exports Wrap Cell Content: 13

cust_null	gender_null	city_null	date_null
0	0	0	0

Result 56 x

Output

Action Output

#	Time	Action	Message	Duration
113	14:44:15	select Order_id, Branch_ID, Order_DateTime from orders LIMIT 0, 50000	20000 row(s) returned	0.000 sec

Object Info Session

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator SQL File 3* data_quality queries Limit to 50000 rows

SCHEMAS

Filter objects

- company_tute
- food_junction
 - branches
 - customers
 - employees
 - menu_items
 - order_details
 - orders
 - reviews
- views
- Stored Procedures
- Functions
- parks_and_recreation
- northfolk_end
- Administration Schemas

Column:
Selling_Price_PKR

Definition:
Selling_Price_PKR int

```

26 -- 1.b
27 • SELECT * from employees;
28 • SELECT
29   SUM(CASE WHEN employees.Branch_id IS NULL OR employees.Branch_id = '' THEN 1 ELSE 0 END) AS branch_null,
30   SUM(CASE WHEN employees.first_name IS NULL OR employees.first_name = '' THEN 1 ELSE 0 END) AS name_null,
31   SUM(CASE WHEN employees.Role IS NULL OR employees.Role = '' THEN 1 ELSE 0 END) AS role_null,
32   SUM(CASE WHEN employees.Hire_Date IS NULL OR employees.Hire_Date = '' THEN 1 ELSE 0 END) AS hired_null,
33   SUM(CASE WHEN employees.Salary_PKR IS NULL OR employees.Salary_PKR = '' THEN 1 ELSE 0 END) AS salary_null
34   from employees;
35

```

Result Grid Filter Rows: Exports Wrap Cell Content: 13

branch_null	name_null	role_null	hired_null	salary_null
0	0	0	0	0

Result 57 x

Output

Action Output

#	Time	Action	Message	Duration
114	14:44:37	SELECT SUM(CASE WHEN customers.first_name IS NULL OR customers.first_name = '' TH...	1 row(s) returned	0.016 sec

Object Info Session

File Edit View Query Database Server Tools Scripting Help

SQL File 3* data_quality queries

Limit to 50000 rows

```

36 -- 1.c
37 * SELECT * from menu_items;
38 * SELECT
39     SUM(CASE WHEN menu_items.Item_Name IS NULL OR menu_items.Item_Name = '' THEN 1 ELSE 0 END) AS item_nu
40     SUM(CASE WHEN menu_items.Category IS NULL OR menu_items.Category = '' THEN 1 ELSE 0 END) AS category_
41     SUM(CASE WHEN menu_items.Selling_Price_PKR IS NULL OR menu_items.Selling_Price_PKR = '' THEN 1 ELSE 0
42     SUM(CASE WHEN menu_items.Cost_PKR IS NULL OR menu_items.Cost_PKR = '' THEN 1 ELSE 0 END) AS cost_nul
43     SUM(CASE WHEN menu_items.Is_Active IS NULL OR menu_items.Is_Active = '' THEN 1 ELSE 0 END) AS active_
44     from menu_items;
45

```

Result Grid

item_name_null	category_null	selling_price_null	cost_null	active_null
0	0	0	0	0

Column: Selling_Price_PKR
Definition: Selling_Price_PKR int

Output

Action Output

Message

115 14:45:59 SELECT SUM(CASE WHEN employees.Branch_id IS NULL OR employees.Branch_id = '' THEN 1 ELSE 0 END) AS branch_nu 1 row(s) returned

SQL File 3* data_quality queries

Limit to 50000 rows

```

46 -- 1.d
47 * SELECT * from order_details;
48 * SELECT
49     SUM(CASE WHEN order_details.Order_ID IS NULL OR order_details.Order_ID = '' THEN 1 ELSE 0 END) AS order_r
50     SUM(CASE WHEN order_details.Menu_item_ID IS NULL OR order_details.Menu_item_ID = '' THEN 1 ELSE 0 END) AS
51     SUM(CASE WHEN order_details.quantity IS NULL OR order_details.quantity = '' THEN 1 ELSE 0 END) AS quantit
52     from order_details;
53
54 -- 1.e
55 * SELECT * from orders;

```

Result Grid

order_null	ordetail_nul	quantity_null
0	0	0

Column: Selling_Price_PKR
Definition: Selling_Price_PKR int

Output

Action Output

Message

116 14:46:35 SELECT SUM(CASE WHEN menu_items.Item_Name IS NULL OR menu_items.Item_Name = '' THEN 1 ELSE 0 END) AS item_nu 1 row(s) returned 0.031

SQL File 3* data_quality queries

Limit to 50000 rows

54 -- 1.e
 55 • SELECT * from orders;
 56 • SELECT
 57 SUM(CASE WHEN orders.Order_ID IS NULL OR orders.Order_ID = '' THEN 1 ELSE 0 END) AS id_null,
 58 SUM(CASE WHEN orders.Customer_ID IS NULL OR orders.Customer_ID = '' THEN 1 ELSE 0 END) AS ordetail_null,
 59 SUM(CASE WHEN orders.Branch_ID IS NULL OR orders.Branch_ID = '' THEN 1 ELSE 0 END) AS branch_null,
 60 SUM(CASE WHEN orders.Order_DateTime IS NULL OR orders.Order_DateTime = '' THEN 1 ELSE 0 END) AS datetime_null
 61 SUM(CASE WHEN orders.status IS NULL OR orders.status = '' THEN 1 ELSE 0 END) AS status_null
 62 from orders;
 63

Result Grid

id_null	ordetail_null	branch_null	datetime_null	status_null
0	0	0	0	0

Column:
 Selling_Price_PKR
 Definition:
 Selling_Price_PKR int

Result 60 x

Output

Action Output

SQL File 3* data_quality queries

Limit to 50000 rows

66 -- 2 DUPLICATE DETECTION
 67 -- Full-row duplicates for customers
 68
 69 -- 2.a
 70 • SELECT *, COUNT(*) AS occurs
 71 FROM branches
 72 GROUP BY Branch_ID, Branch_Name, City -- list all/most columns
 73 HAVING COUNT(*) > 1;
 74
 75 -- 2.b

Result Grid

Branch_ID	Branch_Name	City	occurs
-----------	-------------	------	--------

Column:
 Selling_Price_PKR
 Definition:
 Selling_Price_PKR int

Result 61 x

Output

Action Output

#	Time	Action	Message
118	14:47:46	SELECT SUM(CASE WHEN orders.Order_ID IS NULL OR orders.Order_ID = '' THEN 1 ELSE 0 END) AS id_null, SUM(CASE WHEN orders.Customer_ID IS NULL OR orders.Customer_ID = '' THEN 1 ELSE 0 END) AS ordetail_null, SUM(CASE WHEN orders.Branch_ID IS NULL OR orders.Branch_ID = '' THEN 1 ELSE 0 END) AS branch_null, SUM(CASE WHEN orders.Order_DateTime IS NULL OR orders.Order_DateTime = '' THEN 1 ELSE 0 END) AS datetime_null, SUM(CASE WHEN orders.status IS NULL OR orders.status = '' THEN 1 ELSE 0 END) AS status_null FROM orders;	1 row(s) returned

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SQL File 3* x data_quality queries

Limit to 50000 rows

```

73  HAVING COUNT(*) > 1;
74
75  -- 2.b
76  • SELECT *, COUNT(*) AS occurs
77  FROM customers
78  GROUP BY Customer_ID, First_Name, Last_Name, Gender, City, Join_Date -- list all/most columns
79  HAVING COUNT(*) > 1;
80
81  -- 2.c
82  • SELECT *, COUNT(*) AS occurs

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Customer_ID	First_Name	Last_Name	Gender	City	Join_Date	occurs
-------------	------------	-----------	--------	------	-----------	--------

Column:
Selling_Price_PKR

Definition:
Selling_Price_PKR int.

SQL File 3* x data_quality queries

Limit to 50000 rows

```

110  Having COUNT(*) > 1;
111
112  -- 3.a CHECKING encoding replacement characters, common in csv
113  • SELECT * FROM branches
114  WHERE Branch_ID LIKE '%Ã,â€,??%' OR Branch_Name LIKE '%Ã,â€,??%' OR City LIKE '%Ã,â€,??%';
115
116  -- 3.b
117  • SELECT * FROM customers
118  WHERE Customer_ID LIKE '%Ã,â€,??%' OR First_Name LIKE '%Ã,â€,??%' OR Last_Name LIKE '%Ã,â€,??%';
119

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Branch_ID	Branch_Name	City
-----------	-------------	------

Column:
Selling_Price_PKR

Definition:
Selling_Price_PKR int.

SQL File 3* x data_quality queries

Limit to 50000 rows

```

115
116  -- 3.b
117  • SELECT * FROM customers
118  WHERE Customer_ID LIKE '%Ã,â€,??%' OR First_Name LIKE '%Ã,â€,??%' OR Last_Name LIKE '%Ã,â€,??%' OR Gender LIKE '%Ã,â€,??%';
119
120  -- 3.c
121  • SELECT * FROM employees
122  WHERE Employee_ID LIKE '%Ã,â€,??%' OR Branch_ID LIKE '%Ã,â€,??%' OR First_Name LIKE '%Ã,â€,??%' OR Last_Name LIKE '%Ã,â€,??%';
123
124  -- 3.d

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Customer_ID	First_Name	Last_Name	Gender	City	Join_Date
-------------	------------	-----------	--------	------	-----------

customers 64 x

SQL File 3* x data_quality queries

Limit to 50000 rows

138
139
140 -- 4.a Inconsistent casing/spelling
141 • SELECT DISTINCT City FROM customers
142 ORDER BY City;
143
144 -- 4.b
145 • SELECT DISTINCT Role from employees
146 order by Role
147 ;

Result Grid

City
Faisalabad
Gujranwala
Islamabad
Karachi
Lahore
Rawalpindi
Sialkot

customers 65 x

Output

Action Output

#	Time	Action	Message
122	14:51:14	SELECT * FROM customers WHERE Customer_ID LIKE '%Ã&%;?%' OR First_Name LIKE '...	0 row(s) returned

Object Info Session

Column: Selling_Price_PKR
Definition: Selling_Price_PKR int

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SQL File 3* x data_quality queries

Limit to 50000 rows

139
140 -- 4.a Inconsistent casing/spelling
141 • SELECT DISTINCT City FROM customers
142 ORDER BY City;
143
144 -- 4.b
145 • SELECT DISTINCT Role from employees
146 order by Role
147 ;
148

Result Grid

Role
Branch Manager
Cashier
Chef
Cleaner
Server

employees 66 x

Output

Action Output

#	Time	Action	Message
123	14:53:00	SELECT DISTINCT City FROM customers ORDER BY City LIMIT 0, 50000	7 row(s) returned

Object Info Session

Column: Selling_Price_PKR
Definition: Selling_Price_PKR int

